

Krish Bharal

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EDUCATION

The University of Texas at Dallas

Bachelor of Science in Computer Engineering — GPA: 3.8/4.0

Honors: Academic Excellence Scholar

Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Discrete Mathematics, Probability & Statistics, Digital Circuits

Richardson, TX

Aug 2024 – May 2027

EXPERIENCE

Software Developer

June 2025 – Aug 2025

Care Impact

Leander, TX

- Designed and deployed an efficient Python-based web scraper using Selenium to automate retrieval of patient medical records from a client portal, supporting device recommendation workflows.
- Migrated 5,000+ patient records into Care Impact's databases in less than 2 hours.
- Saved \$10,000+ in costs by creating a reusable data retrieval program.

Full-Stack Developer – PianoWizards (Hacklahoma 2026) | *WebSockets, midi.js*

Feb 2026 – Feb 2026

- Won **1st Place** by architecting and deploying a real-time multiplayer piano game with isolated room management, synchronized client-server state, and low-latency WebSockets communication, delivering a polished production-ready system.
- Designed a fault-tolerant turn-based game engine featuring custom melody validation (± 300 ms timing tolerance), backend state transitions, helping earn more than \$1,200 in competitive prize awards.

Software Engineer – Hand Gesture Recognition Platform | *MediaPipe, FastAPI, Rust*

May 2026 – Present

- Developed a real-time hand gesture recognition platform using Python and MediaPipe to detect, track, and classify one- and two-hand gestures from live webcam input.
- Built a machine learning data pipeline using Pandas and PostgreSQL to collect, label, store, and manage gesture training datasets for custom gesture classification models.
- Designing a cross-platform desktop application using Rust, Dioxus, and FastAPI with secure encrypted credential management and automated update delivery.

Software Developer – Chess Bots | *TypeScript, C++, React, WebSockets*

Aug 2025 – Present

- Built backend command-routing logic for autonomous chess robots using TypeScript and WebSockets, reducing command conflicts and improving real-time movement reliability.
- Created game-state and timeout logic to prevent robot movement conflicts and invalid piece collisions.
- Implemented pathfinding and route optimization algorithms to coordinate autonomous robotic movement while preventing collisions and minimizing execution latency.

Full-Stack Software Developer - krish544.com | *React, Vite, Cloudflare, PostgreSQL*

Nov 2024 – Present

- Designed and deployed a production full-stack web application using React, Vite, Cloudflare Pages Functions, PostgreSQL, and Supabase serving interactive applications and authenticated user content.
- Developed multiple web applications including media processing tools, real-time games, API integrations, and data-driven utilities using React and TypeScript.
- Implemented backend features using Cloudflare Pages Functions, Supabase/PostgreSQL, Drizzle ORM, and WebAuthn passkey authentication to support user login, comments, username changes, and page-specific comment storage.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, Rust, C, C++, SQL, Bash, PowerShell, HTML, CSS

Frameworks & Libraries: React, Node.js, Express, Flask, Next.js, WebSockets, Tailwind CSS, Vite, OpenCV, TensorFlow, FastAPI, WebAuthn, Selenium, React Router, Drizzle ORM, MoviePy, Dioxus

Developer Tools: PostgreSQL, Git, Docker, Google Cloud Platform, VS Code, MATLAB, Eclipse, SolidWorks, Cloudflare, Cloudflare Pages, Supabase, Linux

Concepts: Full-Stack Development, Backend Development, Authentication, Automation, Data Structures, Algorithms, Database Design, Passkeys, Real-Time Systems, CI/CD, REST APIs, Relational Databases, SQL Query Optimization, Cloud Computing, Serverless Architecture, Object-Oriented Programming, Version Control, Agile Development, Unit Testing, Integration Testing, Debugging